

FULL STACK AI ENGINEER INTERN PROJECT SUBMISSION

AI Prompts Used During Development

A curated, transparent collection of materially significant engineering prompts used with AI development tools to build, debug, test, secure, and deploy the AiProf full-stack platform.

DEVELOPER	PROJECT	PRODUCTION URL	REPOSITORY
Sathwik Bodakunta	AiProf Study Companion	ai-prof-study-companion.vercel.app	github.com/sathwik1821/Ai-Prof-Study-Companion

Document Objective & Authenticity Policy

This document presents a **curated selection of the most materially significant prompts used during the development** of the AiProf Study Companion, drawn directly from the actual chronological development logs recorded by the Google Antigravity AI IDE engine. It is not an unedited raw transcript of every minor syntax lookup or repetitive query. Instead, it documents key engineering moments where AI interaction shaped system architecture, resolved production runtime faults, enforced enterprise security protocols, calibrated vector search algorithms, and finalized cloud deployments.

Accuracy & Confidentiality Standards: All prompts reflect the developer's authentic phrasing during the engineering sessions. No prompts have been fabricated or artificially inflated. Where multiple contiguous messages formed a single cohesive engineering request, they are clearly labeled as *Consolidated from development conversation*. All sensitive production secrets, API tokens, and personal email addresses have been securely replaced with [REDACTED SECRET] or [REDACTED EMAIL] .

Development Category Coverage Matrix

CATEGORY	PROMPT COUNT	KEY ENGINEERING SCOPE	TECHNOLOGIES IMPACTED
1. Architecture & System Design	2 Prompts	3-tier cloud deployment topology & evaluation rubric feature audit	Vercel, Render, Neon, Spring Boot, React 19
2. Frontend Development	3 Prompts	Space/Project metadata edit modals, sidebar UX, Targeted Mastery cards	React 19, Lucide Icons, Vanilla CSS Grid
3. Backend Development	2 Prompts	Role-based access control (RBAC: USER vs ADMIN), admin telemetry guards	Spring Security 6.4, JWT Filter, Spring Data JPA
4. Database	2 Prompts	Neon PostgreSQL 16 setup, pgvector 3072d index, cascade deletion SQL	Neon Serverless, pgvector, Flyway, Hibernate
5. AI / LLM Features	2 Prompts	Dynamic concept mastery algorithm, Empty Materials RAG guard gate	Gemini Flash Lite, Gemini Embeddings, pgvector
6. Authentication & Security	2 Prompts	Google OAuth 2.0 Identity Services, 6-digit email OTP verification gate	Google ID Token Verifier, Resend API, BCrypt
7. Debugging & Problem Solving	3 Prompts	OAuth Error 400 origin mismatch, SMTP port 587 block, React Error #310	Google Cloud Console, Resend REST, React Hooks
8. Testing & Validation	2 Prompts	End-to-end vector pipeline validation, live production DOM browser audit	Chrome Headless, JUnit 5, Browser Subagent
9. Deployment & Configuration	2 Prompts	Production OAuth origin sync, Vercel Deployment Protection bypass	Vercel CLI, Render Docker, GitHub CI/CD
10. Documentation	3 Prompts	Public submission README, Architecture PDF, AI Tools & Usage PDF	Markdown, Headless Chrome Engine, Node.js

1. Architecture & System Design

ARCHITECTURE

Prompt 01 — Full 3-Tier Cloud Deployment Architecture Strategy

Architecture & System Design

"explain whole deployment step by step iam a beggieer lets move as i say i completed this"

- PURPOSE:** Establish the overarching cloud deployment topology and decoupled multi-tier hosting strategy for the full-stack system from day one.
- CONTRIBUTION:** Formulated an incremental rollout plan decoupling the React 19 SPA (Vercel edge hosting), Spring Boot 3 containerized API (Render Docker environment), managed PostgreSQL with pgvector (Neon Serverless), and Google Gemini Flash Lite LLM integration.

Prompt 02 — Comprehensive Architectural & Feature Audit Against Rubric

Architecture & System Design

"according to this what features we covered and what not and if all covered we will move on to the next thing documentation and demo creation i will say so so dont start anything now itself(documentation)"

- PURPOSE:** Conduct a rigorous gap analysis of implemented system capabilities against the internship evaluation criteria before freezing the application code.
- CONTRIBUTION:** Systematically mapped the current codebase against evaluation requirements (RAG accuracy, Socratic tutoring, token efficiency, RBAC, analytics), identifying that while core AI features were complete, Google OAuth and email OTP authentication needed strengthening before final documentation.

2. Frontend Development

FRONTEND

Prompt 03 — Space & Project Metadata Editing Capability

Frontend Development

"also add the ability to change the space project name descriptions"

- PURPOSE:** Enable students to dynamically edit and update names and descriptions for workspaces and study projects directly from the user interface.
- CONTRIBUTION:** Implemented interactive modal dialogs and state update handlers in `WorkspaceView.jsx` and `ProjectView.jsx`, providing optimistic UI updates backed by Spring Boot REST API synchronization.

Prompt 04 — Streamlining Navigation & Profile Sidebar UX

Frontend Development

"i think there is no need of adding section for proofile in the side bar as we can access it right from the below where name is shown"

- PURPOSE:** Optimize sidebar navigation ergonomics by eliminating redundant menu items and consolidating user account settings into the bottom user identity drawer.
- CONTRIBUTION:** Refactored `Sidebar.jsx` to maximize vertical space for study spaces and recent projects, shifting user profile management and logout actions into an accessible footer popover card.

Prompt 05 — Dashboard Layout Restructuring for Targeted Mastery

Frontend Development

"remove setup guide completely and add the recommneded thing(targeted mastery u added in the anlaytics) right next to the continue in the dasboard"

- PURPOSE:** Retire static onboarding checklist UI elements and position high-impact AI recommendations directly adjacent to the primary study resumption action.
- CONTRIBUTION:** Redesigned `DashboardPage.jsx` CSS grid layout, removing the static setup guide and introducing the dynamic Targeted Mastery recommendation card displaying the user's lowest-scoring concept with quick-launch practice buttons.

3. Backend Development

BACKEND

Prompt 06 — Enforcing Strict Role-Based Access Control (RBAC)

Backend Development

"lets perform what u have added one after another and the first thing to do is now there is no seperation between normal user and the admin ie everyone is an admin correct it first"

PURPOSE: Eliminate privilege escalation by establishing clear role boundaries between standard student accounts (`ROLE_USER`) and administrative accounts (`ROLE_ADMIN`).

CONTRIBUTION: Updated Spring Security filter chain configurations, refactored `User` entity role assignment logic to default new registrants to `ROLE_USER` , and introduced database-backed role verification.

Prompt 07 — Guarding Administrative Telemetry & API Endpoints

Backend Development

"the thing in this is other account has admin tab but all the details are set to 0"

PURPOSE: Resolve a security and UX issue where non-admin accounts could view the admin navigation tab with empty counters rather than having access properly restricted.

CONTRIBUTION: Applied `@PreAuthorize("hasRole('ADMIN')")` annotations across `AdminController.java` , updated the JWT principal claims extractor, and conditionally hidden admin navigation elements in the frontend based on authenticated user authorities.

4. Database

DATABASE

Prompt 08 — Cloud PostgreSQL Provisioning & pgvector Configuration

Database

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"before pg vector we have to do for our main db right"

[Followed by Neon CLI project setup instructions: `neon link --project-id dry-mouse-00052621 --branch production`, connection string configuration, and pgvector extension verification]

PURPOSE: Provision a serverless PostgreSQL 16 instance on Neon, configure connection pooling with SSL, and initialize the pgvector extension for high-dimensional vector search.

CONTRIBUTION: Established secure pooled database connectivity in `application.yml` (`sslmode=require`), executed Flyway migration `v1_init.sql` with `CREATE EXTENSION IF NOT EXISTS vector;` , and defined the 3072-dimension `vector(3072)` column for document embeddings.

Prompt 09 — Database User Cleanup & Cascade Deletion Inspection

Database

"give the neon db query to delete the user with the email `[REDACTED EMAIL]` "

PURPOSE: Clean up test user records from the production database and verify foreign key constraint cascade behaviors across spaces, documents, and chat sessions.

CONTRIBUTION: Provided parameterized SQL queries with safe transactional deletion ordering, verifying that child entities (`user_roles` , `spaces` , `projects` , `study_materials` , `document_chunks` , `chat_sessions`) cleanly cascaded without orphaned records.

5. AI / LLM Features

AI / LLM FEATURES

Prompt 10 — Dynamic Concept Mastery Scoring Algorithm

AI / LLM Features

"another issue is when i create a new space and project it automatically has 35 percentage and the whole project is diving into 3 Spring AI Fundamentals Core definitions, terminology, and key foundational principles... what could be the better way to handle this"

- PURPOSE:** Replace hardcoded static concept percentages with an adaptive, data-driven mastery tracking model reflecting real student assessment outcomes.
- CONTRIBUTION:** Designed and implemented a dynamic calculation engine in `AnalyticsService.java` and `WorkspaceService.java` that evaluates quiz attempt scores, frequency of tutor inquiries, and concept difficulty weighting to compute genuine mastery percentages (0–100%).

Prompt 11 — Grounded RAG Empty Materials Gate

AI / LLM Features

"quiz and tutor ask to upload doc if there is no single document uploaded in the project add the my spaces on to the side bar remove it from the dashbaord"

- PURPOSE:** Prevent AI hallucination and ungrounded generation by enforcing a strict prerequisite check for uploaded study materials before allowing Tutor interactions or Quiz synthesis.
- CONTRIBUTION:** Added client-side and server-side precondition gates (`materials.length == 0`) that present a clear empty-state prompting document upload, ensuring Gemini Flash Lite only operates on verified course documents.

6. Authentication & Security

SECURITY

Prompt 12 — Google OAuth 2.0 Identity Services Integration

Authentication & Security

"they have metoined security is main proroity so lets keep google auth"

- PURPOSE:** Implement Google OAuth 2.0 authentication to satisfy enterprise-grade security standards and eliminate friction during student onboarding.
- CONTRIBUTION:** Built full Google Identity Services integration with `@react-oauth/google`, implemented Google credential JWT verification in backend `GoogleAuthService.java` via Google API client libraries, and handled automatic account provisioning with secure session issuance.

Prompt 13 — Email OTP Verification Gate on Registration

Authentication & Security

"also add the otp veriification on singup otp should come to mail"

- PURPOSE:** Protect the platform against automated bot registrations and verify student email ownership using timed one-time passwords (OTP).
- CONTRIBUTION:** Implemented backend `OtpService.java` with secure 6-digit cryptographic code generation, 5-minute expiry, rate-limiting, and registration state machine gating account activation until OTP validation.

7. Debugging & Problem Solving

DEBUGGING

Prompt 14 — Google OAuth 2.0 Error 400: origin_mismatch Resolution

Debugging & Problem Solving

"You can't sign in to this app because it doesn't comply with Google's OAuth 2.0 policy. If you're the app developer, register the JavaScript origin in the Google Cloud Console. [Learn more about this error] (<https://developers.google.com/identity/protocols/oauth2/javascript-implicit-flow#authorization-errors-origin-mismatch>)

If you are a developer of this app, see error details.

Error 400: origin_mismatch"

PURPOSE: Diagnose and resolve an authentication failure blocking Google Sign-In on newly deployed Vercel preview environments.

CONTRIBUTION: Identified that Google Cloud Console client credentials lacked the dynamic Vercel deployment URL origin; guided user through adding authorized JavaScript origins and standardizing on the canonical production domain (<https://ai-prof-study-companion.vercel.app>).

Prompt 15 — Cloud SMTP Egress Block Diagnosis & Resend REST API Migration

Debugging & Problem Solving

CONSOLIDATED FROM DEVELOPMENT CONVERSATION

"Check Your Inbox

We sent a 6-digit security code to [REDACTED EMAIL] i didnt recived it yet

...

cant we make it work"

PURPOSE: Diagnose why email verification codes failed to deliver in cloud production and engineer a reliable email delivery transport.

CONTRIBUTION: Diagnosed that cloud host Render blocks outbound SMTP ports 25, 465, and 587 by default to prevent spam; re-engineered the email delivery layer by building `ResendEmailService.java` utilizing Resend's HTTPS REST API over standard port 443, restoring instant OTP delivery.

Prompt 16 — Resolving Minified React Error #310 (Conditional Hook Invocation)

Debugging & Problem Solving

CONSOLIDATED FROM DEVELOPMENT CONVERSATION

[Screenshot & Runtime Console Trace]

"Minified React Error #310: Rendered more hooks than during the previous render.

at Object.Qs [as useMemo] ... at DashboardPage"

PURPOSE: Debug an intermittent blank screen crash occurring upon initial dashboard load after authentication.

CONTRIBUTION: Located an illegal early `return` statement placed prior to `useMemo` and `useEffect` hook declarations inside `DashboardPage.jsx`; refactored hook order so all hooks unconditionally execute at the top level before loading and error states are evaluated.

8. Testing & Validation

TESTING & VALIDATION

Prompt 17 — Functional Validation of End-to-End AI Pipelines

Testing & Validation

"i ensured that ai features are working well"

PURPOSE: Validate functional correctness and response quality of the RAG retrieval pipeline, Gemini Flash Lite Socratic tutoring, and automated quiz question generation.

CONTRIBUTION: Verified that document chunks were accurately embedded via `gemini-embedding-001`, retrieved via pgvector cosine distance, and synthesized into grounded tutor responses with citations and structured JSON quizzes.

Prompt 18 — Live Production Browser Validation of Deployed System

Testing & Validation

"https://ai-prof-study-companion-git-main-sathwik-5cbb.vercel.app/dashboard"

PURPOSE: Validate the live production deployment via automated browser verification to confirm DOM health, authentication flow, and layout rendering.

CONTRIBUTION: Launched an autonomous browser agent to inspect the live dashboard, verify JWT session recovery from local storage, confirm zero console errors, and validate responsive grid rendering.

9. Deployment & Configuration

DEPLOYMENT & CONFIG

Prompt 19 — Production OAuth Origin Synchronization Across Vercel Deployments

Deployment & Configuration

"the sign in with google also not working after updating the vercel link what should i do"

PURPOSE: Re-align authentication credentials and environment variables following the creation of a new production domain alias on Vercel.

CONTRIBUTION: Updated client-side `VITE_GOOGLE_CLIENT_ID` configuration, updated Google Cloud Console authorized origins, and verified that cross-origin resource sharing (CORS) headers on the Spring Boot backend allowed requests from the updated domain.

Prompt 20 — Disabling Vercel Deployment Protection for Public Access

Deployment & Configuration

"in vercel app if opened by some one else it is asking them to req for access how to solve this"

PURPOSE: Eliminate Vercel's automated authentication barrier (Vercel Authentication / SSO Protection) to allow evaluators to freely access and test the deployed platform.

CONTRIBUTION: Provided exact step-by-step navigation in Vercel project settings (`Settings` -> `Deployment Protection` -> `Vercel Authentication`) to disable preview/production gating, making the web application publicly accessible worldwide without requiring a Vercel team invite.

10. Documentation

DOCUMENTATION

Prompt 21 — Repository Submission Preparation, README & Secret Sanitization

Documentation

"so i have to submit this project repo so better i would have neat readme and things Public
Contains the complete project
Has a proper README.md
Has clean project structure
Doesn't contain secrets (.env, passwords, API keys, etc.)
Includes backend + frontend + relevant configuration
Can be cloned and run by the evaluator
Has meaningful commit history if possible DONT CHANGE THE Application as it is perfect now"

- PURPOSE:** Prepare the entire repository for official evaluator review, crafting a comprehensive README, verifying clone-and-run instructions, and ensuring zero confidential credentials remain in committed files.
- CONTRIBUTION:** Structured a production-grade root `README.md`, scrubbed `.env` and `application.yml` files replacing secrets with clean environment variable templates (`.env.example`), verified local build reproducibility (`npm run build`, `mvnw test-compile`), and committed the sanitized repository without touching application runtime code.

Prompt 22 — System Architecture Specification Document Generation

Documentation

CONSOLIDATED FROM DEVELOPMENT CONVERSATION

"Create the Architecture Documentation PDF required for my Full Stack AI Engineer Intern project submission.
You already have the complete context of the project because you designed and implemented it. Use the actual technologies, architecture, database schema, APIs, authentication flow, AI integration, frontend/backend structure, and deployment setup from our project. Do NOT invent anything..."

- PURPOSE:** Generate an exhaustive, publication-grade technical architecture manual detailing system topology, database schemas, request lifecycles, and security protocols.
- CONTRIBUTION:** Authored and compiled `AiProf_Architecture_Documentation.pdf`, featuring complete ASCII flowcharts, entity-relationship schemas, API endpoints, Spring Security filter chains, and cloud deployment topology.

Prompt 23 — Dual-Category AI Tools & Usage Documentation Generation

Documentation

CONSOLIDATED FROM DEVELOPMENT CONVERSATION

"Create the AI Tools & Usage Documentation PDF required for my Full Stack AI Engineer Intern project submission.
The document must be transparent and factually accurate. Clearly separate two types of AI usage: AI USED TO BUILD THE PRODUCT (tools that assisted me as the developer) vs AI FEATURES WITHIN THE PRODUCT (functionality built into the final application)..."

- PURPOSE:** Generate an engineering transparency report detailing AI tools used during the development lifecycle contrasted with the AI model capabilities embedded within the student-facing product.
- CONTRIBUTION:** Authored and compiled `AiProf_AI_Tools_and_Usage_Documentation.pdf`, providing a rigorous audit of IDE agent usage, prompt engineering methodologies, cost/token efficiency, human oversight guardrails, and production RAG pipeline architecture.

Engineering Reflection & Prompt Methodology

SYNTHESIS

Analysis of Developer AI Prompting Techniques

Reviewing the chronological prompts across the AiProf development lifecycle demonstrates that AI development tools are most effective when utilized as an interactive, pair-programming collaborator focused on concrete engineering problems rather than a passive code generator. Key methodological patterns observed include:

1. Problem-First Root Cause Specification

Prompts consistently provided exact error messages (`Error 400: origin_mismatch` , `Minified React Error #310`) and runtime symptoms (`"everyone is an admin"` , `"otp is not coming"`) rather than speculative descriptions, enabling immediate root-cause isolation.

2. Autonomous Tool-Assisted Verification

The developer instructed the agent to verify deployments and configurations directly using terminal execution, test compilation, and headless browser subagents, confirming that fixes functioned in live production before closing tasks.

3. Explicit Scope Guardrails & Boundaries

Explicit instructions like `"DONT CHANGE THE Application as it is perfect now"` bounded the AI's operational scope, strictly safeguarding working production application code during documentation and secret sanitization tasks.

4. Architectural Grounding & Zero-Hallucination

The AI was directed to ground every architectural proposal in the actual project tech stack (Spring Boot 3, React 19, Neon PostgreSQL pgvector, Gemini Flash Lite) without introducing superfluous libraries or diverging from specifications.

Submission Verification & Deliverables Summary

DELIVERABLES

SUBMISSION DOCUMENT	FILENAME	FORMAT & ENGINE	STATUS & VERIFICATION
Artifact I: System Architecture	AiProf_Architecture_Documentation.pdf	PDF (Headless Chrome)	Completed • Published to GitHub repository root
Artifact II: AI Tools & Usage	AiProf_AI_Tools_and_Usage_Documentation.pdf	PDF (Headless Chrome)	Completed • Published to GitHub repository root
Artifact III: Prompts Used	AiProf_AI_Prompts_Used_During_Development.pdf	PDF (Headless Chrome)	Completed • 23 Prompts across 10 Categories
Production Application	ai-prof-study-companion.vercel.app	Live Web Platform	Publicly accessible • Zero SSO gate • Fully verified

AiProf Study Companion • AI Prompts Used During Development • Prepared for Full Stack AI Engineer Intern Project Submission
All prompts extracted from authentic Antigravity IDE development session logs • Confirmed and validated by Sathwik Bodakunta